

Bummps Application – Consolidated Technical Report

Complete technical overview, architecture, feature modules, database models, and API inventory.

Application:	Bummps Mobile App (iOS / Android)	App Version:	1.0.0+13
Framework:	Flutter 3.x (Dart SDK ^3.11.5)	Architecture:	Clean Layered Architecture + GetX
State & DI:	GetX (<code>get: ^4.7.3</code>)	Payment Engine:	Stripe SDK (<code>flutter_stripe: ^14.0.0</code>)
Real-Time Chat:	Socket.IO (<code>socket_io_client: ^3.1.6</code>)	Date Generated:	September 2026

1. Executive Summary

Bummps is a full-featured dating and social networking mobile application developed in Flutter. The platform combines classic swipe-based profile discovery with a community network named **"The Circle"** (featuring member spotlights, topic discussions, and exclusive events), real-time WebSocket messaging, an in-app wallet balance system, and tiered subscription plans processed through **Stripe**.

2. Technology Stack & Key Dependencies

Category	Package / Tool	Version	Role & Function
UI Framework	Flutter SDK	<code>^3.11.5</code>	Cross-platform compiled native mobile client (Android & iOS).
State Management & Routing	<code>get</code>	<code>^4.7.3</code>	Reactive state management (<code>Rx / Obx</code>), dependency injection (<code>Bindings</code>), and route navigation.
Networking	<code>dio</code>	<code>^5.7.0</code>	REST API client with token interception, multipart form data handling, and custom error mapping.
Real-Time Messaging	<code>socket_io_client</code>	<code>^3.1.6</code>	WebSocket communication for live messaging, presence detection, and chat inbox synchronization.
Monetization & Payments	<code>flutter_stripe</code>	<code>^14.0.0</code>	Native Stripe PaymentSheet for credit/debit card subscriptions and wallet balance reloads.
Secure Storage	<code>flutter_secure_storage</code>	<code>^9.2.2</code>	Encrypted storage for JWT bearer access tokens and user identification.
Media Selection	<code>image_picker</code>	<code>^1.1.2</code>	Camera and photo library selection for profile pictures and 6-slot gallery curation.
Icons & Typography	<code>google_fonts</code> , <code>font_awesome_flutter</code>	<code>^8.2.0</code>	Google Fonts typography and FontAwesome vector icons.

3. System Architecture & Project Organization

The project is structured according to clean architecture principles:

- **Core Layer** (`lib/app/core/`): Contains application constants (API routes, Stripe keys), theme definitions (colors, typography), reusable UI widgets, and infrastructure services (DioClient, SocketService, SecureStorageService).
- **Data Layer** (`lib/app/data/`): Contains serialized models (`UserProfile` , `SubscriptionPlan` , `Wallet` , `CircleDashboard`), network providers (`AuthProvider` , `ChatProvider`), and repositories (`AuthRepository` , `ChatRepository`).
- **Modules Layer** (`lib/app/modules/`): Divided into feature-specific domains (Auth, Onboarding, Profile Setup, Home/Discover, Matches, Circle, Messages, Profile, Plans, Help & Support). Each module contains its respective Views, Controllers, and Bindings.
- **Routes Layer** (`lib/app/routes/`): Centralized named route constants (`Routes`) and page registry (`AppPages`).

4. Core Modules & Feature Breakdown

4.1 Onboarding & Authentication

- **Onboarding Flow:** 3-step carousel presenting app concepts with animated page indicators.
- **Authentication Screen:** Segmented toggle allowing users to switch between Login and Registration. Includes email format validation and password visibility toggling.
- **Token Management:** Auth tokens are stored securely in `FlutterSecureStorage`. The `AuthInterceptor` automatically adds `Authorization: Bearer <token>` to every authorized outgoing request.

4.2 6-Step Profile Setup Wizard

- **Step 1 (Basic Details):** First name, date of birth picker, and current location.
- **Step 2 (About You):** Bio with 300-character limit, gender, match preferences, height, education/school, job title, company, lifestyle tags, and spoken languages.
- **Step 3 (Curate Gallery):** 6-slot photo gallery with cover photo designation and multipart image upload support.
- **Step 4 (Verification):** Selfie/liveness verification screen.
- **Step 5 (Preferences):** Age range sliders, search distance radius, and category-based interest chips.
- **Step 6 (Preview & Submission):** Live card preview before submitting the complete multipart registration payload.

4.3 Home Navigation & The 5 Main Tabs

- **1. Discover Tab:**
 - Swipe Right to *Like*, Swipe Left to *Pass*, and Swipe Up to *Super Like*.
 - Match popup modal with direct shortcut to start messaging upon mutual match.
 - Undo / Rewind button calling `/api/matches/rewind` to bring back the last swiped profile.
 - Filter modal sheet for adjusting age range, distance radius, height, lifestyle, and verified-only status.
 - Profile Boost mode with a 30-minute countdown timer.
- **2. Matches Tab:**
 - List of mutual matches with match percentage scores and direct transition to chat.
 - "Who Liked Me" paywall section displaying blurred profile cards for non-subscribers and unlocked cards for active subscribers.
- **3. Community Tab ("The Circle"):**
 - **Circle Dashboard:** Featured spotlight members, active online counts, and community perks.
 - **Trending Discussions:** Categorized forum topics (Fine Dining, Travel, Art, Tech, Culture) with topic creation bottom sheet.
 - **Circle Events:** Real-world exclusive events with dates, locations, and RSVP actions.
 - **Direct Connect:** Allows sending direct connection requests to spotlight members.
- **4. Messages Tab & Chat Detail:**
 - Chat inbox listing conversations with unread badges, timestamps, and online indicators.

- Real-time message exchange via Socket.IO events (`sendMessage` , `receiveMessage` , `getMessages` , `getUserStatus`).
- Typing indicators, message timestamps, and simulated reply capability for demonstrations.
- **5. Profile Tab & Account Settings:**
 - Profile overview, tier badges (Free Trial / Bummps+ / Bummps Pro), and completion stats.
 - **Edit Profile Screen:** Allows updating bio, occupation, education, lifestyle, and gallery photos.
 - **Liked History Screen:** Searchable log of previously liked profiles.
 - **Help & Support Screen:** Support ticket submission form with category selection.

5. Monetization & Subscription Plans

Plan Tier	Pricing	Swipe Limit	Included Privileges
bummps.	\$9.99/mo (\$5.99/mo annual)	~25 Swipes / day	Basic matching & chat, verified profiles only.
bummps+ (Popular)	\$19.99/mo (\$11.99/mo annual)	~75 Swipes / day	Priority visibility, advanced discovery filters, verified profiles.
bummps Pro	\$29.99/mo (\$17.99/mo annual)	Unlimited Swipes	Passport mode (global matching), 5 monthly Super Bummps, maximum spotlight visibility.

Payment Processing Workflow

- **In-App Wallet:** Users can purchase plans directly using their existing wallet balance.
- **Stripe Card Payments:** Users can purchase plans or add funds to their wallet via Stripe native PaymentSheet. The client initiates intent (`/api/plans/wallet/create-intent` or `/api/plans/subscribe`), presents Stripe sheet, and verifies payment on completion (`/api/plans/subscribe/verify` or `/api/plans/wallet/verify`).
- **Access Gating:** Non-subscribers accessing premium features receive an "Access Suspended" prompt redirecting to the Plans page.

6. Backend REST API Endpoint Inventory

Method	Endpoint Path	Module	Description
POST	/api/auth/login	Auth	Authenticates user and returns JWT token + user ID.
POST	/api/auth/register	Auth	Multipart profile registration and initial photo upload.
GET	/api/auth/profile/:id	Profile	Retrieves full profile data for a specific user ID.
PUT	/api/auth/profile	Profile	Updates user details, bio, and gallery photos.
GET	/api/matches/feed	Discover	Retrieves the paginated discovery feed of profiles.
POST	/api/matches/filter	Discover	Fetches feed filtered by age, distance, height, and tags.
POST	/api/matches/like	Matching	Registers a like swipe and returns match status.
POST	/api/swipes/super-like	Matching	Registers a super-like swipe action.
POST	/api/matches/pass	Matching	Registers a pass (nope) swipe action.
POST	/api/matches/rewind	Matching	Undoes the previous swipe action.
GET	/api/matches/my-matches	Matches	Retrieves confirmed mutual matches.
GET	/api/matches/who-liked-me/filter	Matches	Retrieves received likes with filter parameters.
GET	/api/circle/dashboard	Circle	Fetches community dashboard (spotlights, counts, topics).
GET	/api/circle/events	Circle	Retrieves community RSVP events.
POST	/api/circle/discussions	Circle	Creates a new community discussion topic.
GET	/api/plans/allplans	Plans	Fetches all active subscription packages and pricing.
POST	/api/plans/subscribe	Plans	Purchases plan with wallet or creates Stripe client secret.
POST	/api/plans/subscribe/verify	Plans	Verifies Stripe subscription transaction on backend.
GET	/api/plans/wallet	Wallet	Fetches current in-app USD wallet balance.

Method	Endpoint Path	Module	Description
POST	/api/plans/wallet/create-intent	Wallet	Creates a Stripe top-up PaymentIntent.
POST	/api/auth/help-support	Support	Submits user support ticket with category and message.

7. Summary & Recommendations

- **Code Cleanliness:** The application exhibits structured module boundaries, strong separation of concerns, and clean GetX controllers.
- **Network Robustness:** Unified error handling via `ApiException` and automatic token injection via `AuthInterceptor` provides consistent API communication.
- **Real-Time Reliability:** `SocketService` provides real-time chat with event callbacks, reconnect attempts, and typing state management.